1. Context & Purpose

This coursework assesses your ability to apply a *Total Design* process from user need to concept selection and materials justification. You will (i) formulate a Product Design Specification (PDS); (ii) generate concepts using a morphological chart; (iii) select a concept with a Pugh matrix (controlled convergence); and (iv) carry out a structured materials selection using Ashby indices and documentation. The emphasis is on traceable decisions and measurable criteria aligned to requirements.

2. Brief

Choose a simple everyday product or subsystem (e.g. a kettle body, bicycle handlebar, phone case, lightweight beam element, small spring clip, radome-like protective cover). You will:

T1. Define the Market/User Need and PDS (Requirements Specification).
Identify stakeholders, usage contexts, and measurable performance/constraint targets (function, performance, user/market, cost/manufacture, standards/safety, environment/lifecycle). Explain how each requirement is solution-neutral and traceable to the need.

T2. Concept Generation via a Morphological Chart.
Decompose the function; propose at least 4–6 functions and 3–5 means per function. Combine to create a minimum of 6 distinct concepts (sketches at comparable fidelity).

T3. Concept Screening & Selection using Pugh Matrix.
Derive clear, unambiguous criteria from the PDS. Select a suitable datum and evaluate concepts using + / − / S. Provide tallies, interpret trade-offs, and identify a leading concept; justify iterations if criteria/weighting are refined.

T4. Materials Selection for a Key Load-Bearing (or Function-Critical) Part.
Apply the four-step method: Translation, Screening, Ranking (via indices & charts), and Documentation. Derive a relevant material index (e.g. ties, beams, panels, springs) and show quantitative ranking. Discuss process compatibility, cost, sustainability, and risks. Where appropriate, verify with simple back-of-the-envelope checks (stiffness/strength/thermal).

What to submit (single Word or PDF):

- A *report* (see Section) within 3000–3500 words (excludes tables, figures, references, appendices).
- Appendices: full morphological chart, full Pugh matrix, any supporting calculations and charts.

3. Learning Outcomes Assessed

- Apply a staged design process (concept → embodiment → detail) and embed materials selection.
- Formulate a comprehensive PDS and use it to drive objective concept selection.
- Use Ashby charts/indices to screen and rank materials; connect function, material, process, and shape.
- Communicate engineering decisions traceably and professionally.

4. Technical Guidance (formulas & indices)

Use the following canonical forms where they fit your component's function (derive briefly in your context):

$$m_{\min} \propto \frac{\rho}{E^{1/2}} \quad \Rightarrow \quad \text{Maximise } M_b = \frac{E^{1/2}}{\rho} \quad \text{(light, stiff beams)} \tag{1}$$

$$m_{\min} \propto \frac{\rho}{E^{1/3}} \quad \Rightarrow \quad \text{Maximise } M_p = \frac{E^{1/3}}{\rho} \quad \text{(light, stiff panels)} \tag{2}$$

$$m_{\min} \propto \frac{\rho}{\sigma_f} \quad \Rightarrow \quad \text{Maximise } M_t = \frac{\sigma_f}{\rho} \quad \text{(light, strong ties/flywheel specific strength)} \tag{3}$$

$$W_v^{\max} = \frac{1}{2}\frac{\sigma_f^2}{E} \quad \Rightarrow \quad \text{Maximise } \frac{\sigma_f^2}{E} \text{ or } \frac{\sigma_f^2}{\rho E} \quad \text{(springs: volume or mass efficiency)} \tag{4}$$

If fracture safety is relevant, you may reference:

$$K_{IC} = Y\sigma\sqrt{\pi c} \qquad G_c = \frac{K_{IC}^2}{E} \tag{5}$$

5. Submission notes

To submit your coursework report, you will need to use a word processor for typesetting. Do not include images of handwritten notes or calculations in the report. Instead, convert those notes or calculations into a word processor. You can upload either a Word document or a PDF file through the assignment links in the module's Learning Zone on or before the due date of March 27, 2026. It is important to note that the submission portal will generate similarity reports, so be sure to address any issues regarding similarity before your final submission. Generally, a similarity index of 20-30% may trigger a review for academic plagiarism. The portal allows for

multiple submissions, but only the last submission will be graded, and feedback will be provided according to the marking rubric.

6. Marking Rubric (Total = 100%)

Word counts are *guidance* per metric and help you balance depth and breadth.

Metric	What good looks like	Weight	Guide words
Problem Definition & PDS	Clear user/market need; measurable, solution-neutral PDS spanning function, performance, user/market, cost/manufacture, standards/safety, environment/lifecycle; traceability to need.	15%	500–600
Morphological Chart & Concepts	Functional decomposition (4–6 functions); diverse means; at least 6 coherent concepts; unbiased sketches; early pruning via PDS constraints.	15%	400–500
Pugh Matrix & Decision Rationale	Criteria derived from PDS; sensible datum; consistent scoring (+/ –/S); interpretation across rows/columns; controlled convergence; iteration justified.	20%	500–600
Materials Selection (T→S→R→D)	Proper translation; screening limits; correct index derivation; quantitative ranking; Ashby-style reasoning; documentation on processing, cost, sustainability, risks.	30%	900–1100
Engineering Calculations	Back-of-the-envelope checks (stiffness, strength, thermal) with sensible assumptions, units (SI), and clarity.	10%	250–350
Communication & Presentation	Logical structure, neat figures/tables, professional tone; correct referencing of data sources; within overall word limit.	10%	250–350
	Total	**100%**	**3000–3500**

Table 1: Marking rubric with weighting and indicative word counts per metric.

7. Academic Integrity

Where you use property values, standards, or cost data, cite sources. If you include images, ensure you have the rights or redraw schematics. Provide your own calculations; do not submit AI-generated figures without validation.

8. Exemplar Report (Model Structure)

Title & Abstract (150–200 words)

Example (for a Lightweight Handlebar Beam): This report applies a Total Design workflow to a bicycle handlebar. A PDS is formulated from commuter use cases, followed by systematic concept generation via a morphological chart and controlled convergence using a Pugh matrix. The selected concept prioritises stiffness-to-weight and ergonomic adjustability. Materials selection targets a light, stiff beam index $M_b = E^{1/2}/\rho$; candidates are ranked using screening limits and documentation on processing, cost, and sustainability. Back-of-the-envelope checks verify deflection and stress under a 600 N peak load.

1. Market/User Need and PDS

Need (summary): Urban commuters seek a comfortable, fatigue-resistant handlebar with reduced mass for daily rides and climbs.
Key stakeholders: riders, OEM, service shops, recyclers.

PDS snapshot:

- *Function:* Support steering and rider load; mount controls; resist vibration.
- *Performance targets:* Tip deflection < 6 mm at 600 N; factor of safety ≥ 2 for yield.
- *User/market:* Grip diameters compatible with standard controls; ergonomic sweep options.
- *Cost/manufacture:* Target unit cost band; compatible with tube forming or composite layup; DFMA.
- *Standards/safety:* ISO/EN bicycle component norms; safe edges/finishes.
- *Environment/lifecycle:* Recyclability or low embodied energy preference; corrosion resistance.

2. Concept Generation (Morphological Chart)

Function structure: Generate force → transmit → apply at grips; adjust angle; route cables.
Means (examples):

- *Shaft form:* straight / riser / drop / modular
- *Material system:* Al-alloy tube / CFRP layup / bamboo-laminate
- *Join method:* welded stem / clamped / bonded sleeve
- *Vibration control:* wall tuning / visco insert / layup damping
- *Cable strategy:* external clips / internal routing

Combine systematically to produce 6 concepts with matched-fidelity sketches. Highlight early eliminations using PDS constraints (e.g. incompatible routing or manufacturability).

3. Concept Selection (Pugh Matrix)

Criteria derived from PDS: stiffness, mass, manufacturability, risk, cost band, ergonomics, sustainability. Choose datum (e.g. current OEM aluminium design). Score each concept

(+/−/S) and present totals. Discuss patterns (e.g. CFRP variants trend + in stiffness-to-weight but − in cost/risk). State selected concept and any refinement loop.

4. Materials Selection (T→S→R→D)

Translation: Function: light, stiff beam; Constraints: outer diameter limit, minimum factor of safety; Objective: minimise mass for specified stiffness; Free variables: material, wall thickness.

Index:

$$\text{For beams at prescribed stiffness:} \qquad \text{Maximise } M_b = \frac{E^{1/2}}{\rho}. \tag{6}$$

Screening limits: $E > 30$ GPa, $\sigma_y > 250$ MPa, $\rho < 3$ g/cm^3, suitable process routes (extrusion/layup), environment resistance.

Ranking: Compare (e.g.) 6061-T6-Al, 7000-series Al, Ti-6Al-4V, CFRP (unidirectional + woven), GFRP. Discuss where each sits relative to E-ρ clusters and M_b lines; note that CFRP and selected woods often excel in $E^{1/2}/\rho$, while ceramics are excluded by toughness needs for impact.

Documentation: Summarise process, cost per kg and per m, recyclability, supplier notes, typical failure modes.

Illustrative quick check (cantilever proxy):

$$\delta = \frac{FL^3}{3EI'} \qquad I = \frac{\pi}{64}\left(D^4 - d^4\right) \tag{7}$$

Assume $F = 600$ N, $L = 0.25$ m, choose sizing to meet $\delta < 6$ mm and $\sigma_{max} < \sigma_y/2$; compare CFRP vs. 7000-series Al mass for equal stiffness.

5. Risk, Sustainability & Lifecycle

Identify key risks (impact damage for CFRP; corrosion for Al in winter salts), mitigations (protective sleeves, coatings, QC), and end-of-life routes. Comment on embodied energy vs. use-phase savings.

6. Conclusions

Recap the traceable chain: need → PDS → concepts → Pugh selection → materials index & ranking → verification. State final recommendation and next experimental steps (prototype & test against PDS).

References

List data sources for properties, costs, standards, and any images used. Use a consistent citation style.

Appendix A: Full Morphological Chart

Provide the full matrix and legend.

Appendix B: Full Pugh Matrix

Include the criteria list, datum selection rationale, scoring table, and tallies.

Appendix C: Calculations

Show workings for stiffness/strength checks, units, and assumptions.